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RESEARCH ARTICLE



Adaptive dynamic programming optimal tracking control of multi-UAVs based on zero-sum game

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ABSTRACT

This paper investigates an adaptive dynamic programming (ADP) optimal tracking control algorithm for multi-UAV systems based on zero-sum game theory, addressing external disturbances during flight. By formulating the bounded L_2 gain problem as a two-person zero-sum game between control strategies and external disturbances, the Hamilton-Jacobi-Isaacs (HJI) equation is constructed to derive the Nash equilibrium solution. To overcome the computational challenges of solving the HJI equation, a three-layer neural network structure comprising evaluation, execution, and disturbance networks is employed, integrated with the ADP algorithm to approximate the value function and optimize the control strategy iteratively. Comprehensive simulations demonstrate the proposed method's superior trajectory tracking performance and robustness compared to Sliding Mode Control (SMC). The results confirm the effectiveness of the ADP-based approach in achieving real-time, adaptive control in complex and dynamic multi-UAV environments.

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Multi-UAVs; zero-sum game; adaptive dynamic programming; neural networks

1. Introduction

The application of UAVs has developed rapidly over the past decades. Due to the characteristics of small size, high mobility and low energy consumption, UAVs have been widely used in public, civil and military fields [1,2]. Currently, the research of multi-UAV systems is a hot spot. Each UAV in a multi-UAVs system can be regarded as an independent decision maker. It participates in cooperation and competition in the game. Game theory provides an effective method to obtain the optimal strategy for multi-player game decision-making [3–6].

The development of game theory as a branch of mathematics and economics that studies the interactions of decision makers can be traced back to the early 20th century. In the early days, Zermelo and Von Neumann respectively conducted preliminary studies on game theory, Zermelo proposed the concept of zero-sum games, and von Neumann elaborated the mathematical basis of game theory in his works, laying the foundation for this field. For example, in [7], alternative positioning, navigation, and timing are achieved by utilizing games in satisfaction form and reconfigurable intelligent surfaces. Its core lies in constructing a new game model, which fully considers the influence of the reconfigurable characteristics of the intelligent surface on positioning, navigation, and timing. Different from traditional methods, it is no longer limited to simple path planning or resource allocation games. Instead, it closely combines environmental factors with system

performance and optimizes the spatio-temporal performance of UAVs in complex environments through the dynamic adjustment of the intelligent surface.

With the development of the times, researchers have studied the combination of game theory and the control of UAVs. Initially, the main focus was on the application of game theory in a single UAV system. It explores the optimal decision-making of a single UAV in terms of resource allocation, path planning and so on. With the continuous progress of artificial intelligence and autonomous systems technology, the application of game theory in multi-UAVs has become increasingly complex and in-depth. The game strategy between each UAV in a group is beginning to be paid attention to optimize the overall performance. ADP as the outstanding reinforcement learning-based algorithm [8–10], was first proposed by Werbos in 1977 for developing optimal controllers [11]. In recent years, with the deepening of ADP theory and the improvement of computational power, ADP has begun to be more widely used in UAVs, including path planning [12], resource allocation [13], and system collaboration [14]. There are some results of research on game theory and the application of adaptive dynamic planning algorithms in multi-UAVs [15–20].

In [15], the collaborative combat problem in multi-UAV systems was investigated. The algorithmic subsystem primarily based on game theory is utilized to solve optimal combat strategy problems between opposing forces under certain scenarios. In reference [18], the

self-organizing the problem of self-organized collaboration of multi-UAV systems is studied. The description of multi-UAV system collaboration and optimal control is given in the framework of cooperative game, and the Nash method is used to give the game optimization consistency control algorithm. In reference [19], a multi-UAV cooperative game model is established for the multi-UAV cooperative game problem by considering the objective value function and cost function of the cooperative alliance. Through a cooperative game solving method, the optimal multi-UAV strategy is finally obtained. In reference [17], a game-based distributed control algorithm is proposed for the problem of controlling multiple UAVs in arbitrary formation. Gradient descent is used to realize the solution of Nash equilibrium point. In reference [16], two perspectives of UAV mission planning and formation flying are reviewed. In [20], a distributed game strategy for UAV formation with external disturbances and obstacles is investigated. Although the reference [20] considers potential external disturbances, the adoption of multiple technical approaches may increase the overall complexity of the method, potentially making algorithm implementation and debugging more cumbersome.

The importance of path planning is undoubtedly emphasized when facing external disturbances. Path planning not only affects the trajectory of the UAV system, but also directly relates to how the system avoids disturbances and bypasses obstacles, as well as efficiently accomplishes tasks in complex environments. Among them, adaptive dynamic planning algorithm is a commonly used method in path planning. This method is often used to solve Nash equilibrium solutions for multi-agent systems under game problems [21–26]. In [26], an optimal cooperative fault-tolerant control law based on dynamic programming was proposed in order to address unknown actuator failures during UAV formation flight. Ultimately, high precision control of UAV formation flight is realized. In [21], a target tracking control algorithm based on approximate dynamic programming is proposed. The rollout algorithm is utilized to solve the optimal tracking decision, which realizes the flexible and effective accurate tracking of the game target. In [24], a distributed adaptive optimal control scheme is proposed to solve the nonzero sum differential game problem for UAV systems in formation with input saturation limitation. A single network structure is designed for each UAV to approximate the solution of the coupled Hamilton-Jacobi equations. From the above analysis, it can be seen that there are relatively few studies that combine game theory with ADP algorithms for multi-UAV systems. Although literature [24] has been studied in conjunction with game theory, the single network structure used has shortcomings. Compared to the multi-network structure, the single network structure may limit the expressive power of the model and make it difficult to cover the diversity

of features in the data. When dealing with complex tasks, a single network may not be able to adequately capture information from different levels and domains, thus limiting the overall performance of the model.

In most current research, the focus has predominantly been on game strategies within UAV formations, with relatively little attention given to applying game-theoretic decision-making to the optimal tracking control of multi-UAV systems. Therefore, this paper designs an optimal tracking control algorithm for multi-UAV systems based on game theory and adaptive dynamic programming (ADP). This approach combines the rigour of theoretical game models with the versatility of adaptive algorithms. The core contributions of this work are as follows:

(1) We integrate zero-sum game theory with adaptive dynamic programming (ADP) in multi-UAV systems, establishing an innovative theoretical framework. This framework not only broadens the theoretical discourse but also deepens the analysis of complex systems, enabling a more nuanced understanding of UAV coordination and their dynamic interaction with the environment. Through this theoretical advancement, we enhance the internal coordination mechanisms among UAVs, thereby improving the overall operational efficiency and responsiveness of the system.

(2) We propose a neural network architecture comprising evaluation, execution, and disturbance components to effectively solve the computationally intensive Hamilton-Jacobi-Isaacs (HJI) equation, a key technical challenge in traditional ADP implementations. This architecture significantly improves processing speed and computational efficiency through the collaborative operation of the three networks. Moreover, our ADP algorithm optimizes control strategies through continuous iterative updates, enabling the UAV system to adapt its behaviour in real time when faced with environmental changes and dynamic internal states, thus enhancing adaptability and overall system stability.

(3) Comprehensive simulation experiments validate the proposed method's performance advantages in handling complex and disruptive environments. Additionally, comparative experiments with Sliding Mode Control (SMC) techniques demonstrate that while SMC excels in certain scenarios, our adaptive dynamic programming approach shows more pronounced advantages in maintaining system stability and handling dynamic disturbances. These experiments not only verify the effectiveness of our method but also highlight its unique strategic value for real-time adaptive control systems in addressing complex challenges.

2. Problem description

The communication topology of a multi-UAVs is modelled using the graph $G = \{V, \varsigma, E\}$, where ς denotes a nonempty finite set $V = \{\nu_1, \nu_2, \dots, \nu_N\}$, the set of

edges $\zeta \subseteq V \times V$, and defines the connectivity matrix E make $E = [a_{ij}]$. If $(v_i, v_j) \in \zeta$, then $a_{ij} > 0$, otherwise $a_{ij} = 0$.

A node j is said to be a neighbour of a node i , if node i can send a message to node j , and the set of neighbour nodes of i can be denoted as $N_i = \{j : (j \in V), (i, j) \in \zeta\}$. A graph is said to contain a directed spanning tree if there are nodes in the graph that can reach other nodes via directed paths. Define the incidence matrix D as the diagonal matrix $D = \text{diag}(d_i)$, where $d_i = \sum_{j=1, j \neq i}^N a_{ij}$, denotes the incidence of node i . The graph Laplacian matrix L is defined as $L = D - E$, and the elements of the Laplacian matrix l_{ij} are defined as follows:

$$l_{ij} = \begin{cases} -a_{ij}, & i \neq j, \\ \sum_{m \in N_i} a_{im}, & i = j. \end{cases} \quad (1)$$

In this paper, a system consisting of N UAVs subject to unknown external disturbances is considered. The trajectory tracking problem of the multi-UAVs system is addressed. The dynamics equation of the multi-UAVs system can be expressed as:

$$x_i(k+1) = f(x_i(k)) + g_i u_i(k) + h_i d_i(k) \quad (2)$$

Where $x_i(k)$, $u_i(k)$, $d_i(k)$ denotes the state, the control input, and the external disturbance of node i , respectively. And f , g_i , h_i is a nonlinear function and $f(0) = 0$. For ease of representation we write $x_i(k)$, $u_i(k)$, $d_i(k)$ as x_{ik} , u_{ik} , d_{ik} .

The state of the leader UAV node is $X_{0k} \in R_n$, assuming that it satisfies the dynamics:

$$x_{0(k+1)} = Fx_{0k} \quad (3)$$

To ensure that all UAVs remain synchronized with the leader, a local relative coordination error is designed to describe the coordination goals of the team. The local neighbourhood coordination error for the i_{th} UAV is defined as:

$$\delta_{ik} = \sum_{j \in N_i} a_{ij}(x_{ik} - x_{jk}) + c_i(x_{ik} - x_{0k}) \quad (4)$$

Where $c_i \geq 0$ is the fixed gain at node i . a_{ij} is the coefficient related to the relationship between UAVs. If there is information exchange between UAV i and UAV j , then $a_{ij} = 1$; otherwise, $a_{ij} = 0$.

Neighborhood coordination error for all UAVs:

$$E = \sum_{i=1}^N \delta_{ik} \quad (5)$$

Local neighbour coordination dynamic error of the i_{th} UAV:

$$\delta_{i(k+1)} = F_i(\delta_{ik}, u_{ik}, u_{jk}, d_{ik}, d_{jk})$$

$$\begin{aligned} &= f\delta_{ik} + (d_i + c_i)g_i u_{ik} \\ &\quad - \sum_{j \in N_i} a_{ij}g_j u_{jk} + (d_i + c_i)h_i d_{ik} \\ &\quad - \sum_{j \in N_i} a_{ij}h_j d_{jk} \end{aligned} \quad (6)$$

Here, u_{jk} is the control strategy of UAV i neighbours, and d_{jk} represents external disturbances in the neighbourhood of UAV i . This represents a dynamic system with multiple control inputs and external disturbances.

Considering the multi-UAV formation described by Equations (3.1) and (3.2), where all followers satisfy $d_{ik} \neq 0$, the goal is to find a control strategy u_{ik} , that ensures the closed-loop system gradually stabilizes and satisfies the bounded L_2 gain condition:

$$\begin{aligned} &\sum_{k=0}^{\infty} \left(\delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} + \sum_{j \in N_i} u_{jk}^T R_{ij} u_{jk} \right) \\ &\leq \gamma^2 \left(d_{ik}^T P_{ii} d_{ik} + \sum_{j \in N_i} d_{jk}^T P_{ij} d_{jk} \right) + V_i(\delta_i(0)) \end{aligned} \quad (7)$$

Among them, Q_{ii} , R_{ii} , R_{ij} , P_{ii} , and P_{ij} are all positive definite matrices, $\gamma > 0$ is the coefficient of the non-optimal performance index of UAV i , and V_i is a bounded function, and $V_i(0) = 0$.

3. Multi-UAV zero-sum game strategy

3.1. Definition of performance indicators

Define performance metrics for each UAV:

$$\begin{aligned} &J_i(\{\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}\}) \\ &= \sum_{k=0}^{\infty} M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \\ &= \frac{1}{2} \sum_{k=0}^{\infty} \left(\delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} + \sum_{j \in N_i} u_{jk}^T R_{ij} u_{jk} \right. \\ &\quad \left. - \gamma^2 \sum_{k=0}^{\infty} d_{ik}^T P_{ii} d_{ik} + \sum_{j \in N_i} d_{jk}^T P_{ij} d_{jk} \right) \end{aligned} \quad (8)$$

In the performance index equation (8), δ_{ik} represents the local neighbourhood coordination error of UAV i , which measures the state deviation between UAVs and the leader. It is specifically determined by Equation (4). u_{ik} and u_{-ik} denote the control inputs of UAV i and its neighbouring UAVs, respectively, determined by the optimal control strategy from the zero-sum game. d_{ik} and d_{-ik} represent the external disturbances acting on UAV i and its neighbouring UAVs, respectively, which are determined by the opponent's strategy in the game. Q_{ii} , R_{ii} , R_{ij} , P_{ii} , P_{ij} are positive-definite weighting matrices associated with the costs of errors, control

inputs, and disturbances, selected based on system performance requirements and design specifications. γ is the proportional coefficient of the non-optimal performance index, typically adjusted through experiments or parameter tuning.

At the same time, The value function corresponding to UAV i is given:

$$V_i(\delta_{ik}) = \sum_{n=k}^{\infty} M_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) \quad (9)$$

In the value function Equation (9), the value function V_i represents the cumulative performance index of UAV i , which evaluates the overall performance of the current state. Its instantaneous value M_i is composed of the weighted costs of errors, control inputs, and disturbances. The value function can be obtained by solving the HJI equation or approximated using neural network-based algorithms.

Therefore, the bounded L_2 gain problem can be transformed into the following multi-UAV zero-sum game:

$$V_i(\delta_{ik}) = \min_{u_{ik}} \max_{d_{ik}} M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \quad (10)$$

If saddle point $(\bar{u}_{ik}^*, \bar{d}_{ik}^*)$ exists in the sense of the game, then the game has a unique solution:

$$\begin{aligned} V_i^*(\delta_{ik}) &= \min_{u_{ik}} \max_{d_{ik}} M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \\ &= \max_{d_{ik}} \min_{u_{ik}} M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \end{aligned} \quad (11)$$

Definition 3.1: The UAVs zero-sum game optimal control strategy and external disturbance strategy (u_{ik}^*, d_{ik}^*) for $\forall i \in N$ if the inequality (12) is satisfied, then there is a global Nash equilibrium solution.

$$\begin{aligned} J_i(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}, d_{-ik}^*) &\leq J_i(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \\ &\leq J_i(\delta_{ik}, u_{ik}, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \end{aligned} \quad (12)$$

Therefore, the optimal control and the external disturbance for each UAV i can be expressed as follows, respectively:

$$\begin{cases} u_{ik}^* = -(d_i + c_i)R_{ii}^{-1}g_i^T \nabla V_i^*(\delta_{i(k+1)}) \\ d_{ik}^* = -\frac{1}{\gamma^2}(d_i + c_i)P_{ii}^{-1}h_i^T \nabla V_i^*(\delta_{i(k+1)}) \end{cases} \quad (13)$$

Where, $\nabla V_i^* = \frac{\partial V_i}{\partial \delta_{ik}}$ one can obtaine:

$$\begin{cases} u_{ik}^* = -(d_i + c_i)R_{ii}^{-1}g_i^T \lambda_{i(k+1)}^* \\ d_{ik}^* = -\frac{1}{\gamma^2}(d_i + c_i)P_{ii}^{-1}h_i^T \lambda_{i(k+1)}^* \end{cases} \quad (14)$$

Among them, $\lambda_{ik} = \lambda_i(k)$ is the adjoint variable of the unmanned aerial vehicle (UAV) i . At the same time, the zero-sum game Bellman optimal equation is obtained:

$$V_i^*(\delta_{ik}) = V_i^*(\delta_{i(k+1)}) + M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \quad (15)$$

Thus, the Hamiltonian equation for each UAV i can be defined as:

$$\begin{aligned} H_i(\delta_{ik}, \lambda_{i(k+1)}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \\ = \lambda_{i(k+1)}^T \left(f\delta_{ik} + (d_i + c_i)g_i u_{ik} - \sum_{j \in N_i} a_{ij}g_j u_{jk} \right. \\ \left. + (d_i + c_i)h_i d_{ik} - \sum_{j \in N_i} a_{ij}h_j d_{jk} \right) \\ + \frac{1}{2} \left(\delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} + \sum_{j \in N_i} u_{jk}^T R_{ij} u_{jk} \right. \\ \left. - \gamma^2 \sum_{k=0}^{\infty} d_{ik}^T P_{ii} d_{ik} + \sum_{j \in N_i} d_{jk}^T P_{ij} d_{jk} \right) \end{aligned} \quad (16)$$

Bringing the Equation (14) into (16) yields the HJI equation:

$$\nabla V_i(\delta_{i(k+1)})^T \Pi_i + \Xi_i = 0 \quad (17)$$

Where,

$$\begin{aligned} \Pi_i &= f\delta_{ik} - (d_i + c_i)^2 g_i R_{ii}^{-1} g_i^T \nabla V_i(\delta_{i(k+1)}) \\ &+ \sum_{j \in N_i} a_{ij}(d_j + c_j) g_j R_{jj}^{-1} g_j^T \nabla V_j(\delta_{j(k+1)}) \\ &+ \frac{1}{\gamma^2} (d_i + c_i)^2 h_i P_{ii}^{-1} h_i^T \nabla V_i(\delta_{i(k+1)}) \\ &- \frac{1}{\gamma^2} \sum_{j \in N_i} a_{ij}(d_i + c_i) h_j P_{jj}^{-1} h_j^T \nabla V_j(\delta_{j(k+1)}) \end{aligned} \quad (18)$$

$$\begin{aligned} \Xi_i &= \left(\frac{1}{2} \delta_{ik}^T Q_{ii} \delta_{ik} + \frac{1}{2} (d_i + c_i)^2 \nabla V_i(\delta_{i(k+1)})^T g_i \right. \\ &\quad \left. R_{ii} g_i^T \nabla V_i(\delta_{i(k+1)}) \right) \\ &+ \left(\frac{1}{2} \sum_{j \in N_i} (d_i + c_i)^2 \nabla V_j(\delta_{j(k+1)})^T g_j \right. \\ &\quad \left. R_{jj}^{-1} R_{ij} R_{jj}^{-1} \nabla V_j(\delta_{j(k+1)}) \right) \\ &- \left(\frac{1}{2\gamma^2} (d_i + c_i)^2 \right. \\ &\quad \left. \nabla V_i(\delta_{i(k+1)})^T h_i P_{ii}^{-1} h_i^T \nabla V_i(\delta_{i(k+1)}) \right) \\ &- \left(\frac{1}{2\gamma^2} \sum_{j \in N_i} (d_i + c_i)^2 \nabla V_j(\delta_{j(k+1)})^T h_j P_{jj}^{-1} \right. \\ &\quad \left. \sum_{j \in N_i} P_{ij} P_{jj}^{-1} \nabla V_j(\delta_{j(k+1)}) \right) \end{aligned} \quad (19)$$

Boundary conditions:

$$V_i(0) = 0 \quad (20)$$

Then the above equation can be rewritten as:

$$\begin{cases} H_i(\delta_{ik}, \nabla V_i^*(\delta_{i(k+1)}), u_{ik}^*, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) = 0 \\ V_i(0) = 0 \end{cases} \quad (21)$$

Lemma 3.1: (1) Let $\gamma > \gamma^*$, where γ^* is the optimal value of γ . Assume that $V_i^*(\delta_{ik}) > 0$, ($i \in N$) is the positive – definite solution of the coupled HJI equation (17), and assume that the control strategies of the neighbouring UAVs are all optimal. Then the closed-loop system is asymptotically stable at the equilibrium point $\varepsilon_{i(k+1)}$. (2) When all followers choose their respective optimal control strategies u_{ik}^* , the bounded L_2 -gain condition holds for all disturbances.

Note: The proof of Lemma 3.1 refers to the proof of Theorem 1 in reference [27].

3.2. Stability analysis and Nash equilibrium solution

Theorem 3.1: Let $\gamma > \gamma^*$, assume that Game $V_i(\delta_{ik})$ has finite values and that the strategies of the lead UAVs are all optimal, the optimal performance metrics for UAV i are:

$$J_i^*(\delta_{in}, u_{in}^*, u_{-in}^*, d_{in}^*, d_{-in}^*) = V_i^*(\delta_{in}) \quad (22)$$

Then then all UAVs are in Nash equilibrium.

Proof: It can be obtained from Equation (15) and the definition of the value function:

$$\begin{aligned} J_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) \\ = V_i^*(\delta_{in}) + \sum_{k=n}^{\infty} (M_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) \\ - M_i^*(\delta_{in}, u_{in}^*, u_{-in}^*, d_{in}^*, d_{-in}^*)) \end{aligned} \quad (23)$$

From the definition of value function:

$$\begin{aligned} M_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) \\ - M_i^*(\delta_{in}, u_{in}^*, u_{-in}^*, d_{in}^*, d_{-in}^*) \\ = \frac{1}{2}(u_{ik} - u_{ik}^*)^T R_{ii}(u_{ik} - u_{ik}^*) \\ + (u_{ik}^*)^T R_{ii}(u_{ik} - u_{ik}^*) \\ + \frac{1}{2} \sum_{j \in N_i} (u_{jk} - u_{jk}^*)^T R_{ij}(u_{jk} - u_{jk}^*) \\ + \sum_{i \in N_i} (u_{jk}^*)^T R_{ij}(u_{jk} - u_{jk}^*) \end{aligned}$$

$$\begin{aligned} - \frac{\gamma^2}{2}(d_{ik} - d_{ik}^*)^T P_{ii}(d_{ik} - d_{ik}^*) \\ - \gamma^2 d_{ik}^* P_{ii}(d_{ik} - d_{ik}^*) \\ - \frac{\gamma^2}{2} \sum_{j \in N_i} (d_{jk} - d_{jk}^*)^T P_{ij}(d_{jk} - d_{jk}^*) \\ - \gamma^2 \sum_{j \in N_i} (d_{jk}^*)^T P_{ij}(d_{jk} - d_{jk}^*) \end{aligned} \quad (24)$$

Bringing the Equation (24) into (22), yields:

$$\begin{aligned} J_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) = V_i^*(\delta_{in}) \\ + \sum_{k=n}^{\infty} \left(\frac{1}{2}(u_{ik} - u_{ik}^*)^T R_{ii}(u_{ik} - u_{ik}^*) + (u_{ik}^*)^T R_{ii}(u_{ik} - u_{ik}^*) \right. \\ + \frac{1}{2} \sum_{j \in N_i} (u_{jk} - u_{jk}^*)^T R_{ij}(u_{jk} - u_{jk}^*) + \sum_{i \in N_i} (u_{jk}^*)^T R_{ij}(u_{jk} - u_{jk}^*) \\ - \frac{\gamma^2}{2}(d_{ik} - d_{ik}^*)^T P_{ii}(d_{ik} - d_{ik}^*) - \gamma^2 d_{ik}^* P_{ii}(d_{ik} - d_{ik}^*) \\ - \frac{\gamma^2}{2} \sum_{j \in N_i} (d_{jk} - d_{jk}^*)^T P_{ij}(d_{jk} - d_{jk}^*) \\ \left. - \gamma^2 \sum_{j \in N_i} (d_{jk}^*)^T P_{ij}(d_{jk} - d_{jk}^*) \right) \end{aligned} \quad (25)$$

The optimal performance index J_i^* can be obtained from (25), such that:

$$J_i^*(\delta_{in}, u_{in}^*, u_{-in}^*, d_{in}^*, d_{-in}^*) = V_i^*(\delta_{in}) \quad (26)$$

According to (23), from the definition of the value function, one can get:

$$\sum_{k=n}^{\infty} (M_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) - M_i^*(\delta_{in}, u_{in}^*, u_{-in}^*, d_{in}^*, d_{-in}^*)) \geq 0 \quad (27)$$

Therefore,

$$J_i(\delta_{in}, u_{in}, u_{-in}, d_{in}, d_{-in}) \geq J_i^*(\delta_{in}, u_{in}^*, u_{-in}^*, d_{in}^*, d_{-in}^*) \quad (28)$$

From the Equation (28),

$$J_i(\delta_{ik}, u_{ik}, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \geq J_i^*(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \quad (29)$$

The same reasoning proves that,

$$J_i(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}, d_{-ik}^*) \leq J_i^*(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \quad (30)$$

Summarizing, it can be concluded that:

$$\begin{aligned} J_i(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}, d_{-ik}^*) &\leq J_i^*(\delta_{ik}, u_{ik}^*, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \\ &\leq J_i(\delta_{ik}, u_{ik}, u_{-ik}^*, d_{ik}^*, d_{-ik}^*) \end{aligned} \quad (31)$$

The proof of Theorem 3.1 is completed. ■

4. An ADP algorithm based on zero-sum games

In this section, the objective is to find the optimal solutions u_{ik}^* and d_{ik}^* for the two-player zero-sum game problem in multi-UAVs system. The ADP iterative algorithm is used by iterating until convergence to the optimal solution.

4.1. Derivation of the ADP algorithm based on zero-sum game

First, an initial-value function $V_{(0)k}$ is selected, and the value of this initial-value function is not necessarily the optimal value for the function, set $\gamma > 0$. Then $V_{(i)k}$ is updated as shown in Equation (32):

$$V_{(i+1)k}(\delta_{ik}) = \min_{u_{ik}} \max_{d_{ik}} \left\{ \delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} - \gamma^2 d_{ik}^T P_{ii} d_{ik} + V_i(\delta_{i(k+1)}) \right\} \quad (32)$$

The iterative strategy is updated as follows:

$$\begin{cases} u_{ik} = -(d_i + c_i) R_{ii}^{-1} g_i^T \lambda_{i(k+1)}^* \\ d_{ik} = -\frac{1}{\gamma^2} (d_i + c_i) P_{ii}^{-1} h_i^T \lambda_{i(k+1)}^* \end{cases} \quad (33)$$

Secondly, substitute the updated Equation (33) into (32) to obtain Equation (34), and calculate the value of $V_{(1)k}(\delta_{ik})$ when $i = 0$ through Equation (34):

$$V_{(i+1)k}(\delta_{ik}) = \delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} - \gamma^2 d_{ik}^T P_{ii} d_{ik} + V_{ik}(\delta_{i(k+1)}) \quad (34)$$

Thus, $V_1(\delta_{ik})$ can be calculated, and for $i = 1, 2, \dots$, the same value iteration process is repeated. In addition, it should be satisfied that $\forall i \geq 0$, $V_i(0) = 0$, where i is the iteration index and k is the state index. As a numerical iteration algorithm, this iterative ADP algorithm does not require any initial stabilizing controller.

4.2. Evaluation network design

It is known through the ADP algorithm that nonlinear HJI equations are difficult to solve. Therefore, a three-layer neural network is used to approximate the function values. Next, design the evaluation network.

According to the generalized approximation property of neural networks, the value function can be approximated by a neural network as:

$$V_i(\delta_{i(k+1)}) = W_i^T \psi(\delta_{i(k+1)}) + \eta_{i(k+1)} \quad (35)$$

Where W_i is the weights, $\psi(\delta_{i(k+1)})$ is the vector of activation functions for evaluating the neural network, the number of neurones in the hidden layer of the neural network is h . The neural network approximation error is $\eta_{i(k+1)}$ when $h \rightarrow \infty$ converges to zero consistently.

Assuming that the current weights are estimated as $\hat{W}_i$, the output of the evaluation network is:

$$\hat{V}_i(\delta_{i(k+1)}) = \hat{W}_i^T \psi(\delta_{i(k+1)}) \quad (36)$$

The control strategy and external disturbances are approximated:

$$\begin{cases} \hat{u}_{ik} = -(d_i + c_i) R_{ii}^{-1} g_i^T \nabla \hat{V}_i(\delta_{i(k+1)}) \\ \hat{d}_{ik} = \frac{1}{\gamma^2} (d_i + c_i) P_{ii}^{-1} h_i^T \nabla \hat{V}_i(\delta_{i(k+1)}) \end{cases} \quad (37)$$

The residuals generated by the neural network are denoted as:

$$\begin{aligned} e_{H_i} = \frac{1}{2} & \left(\delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} + \sum_{j \in N_i} u_{jk}^T R_{ij} u_{jk} \right. \\ & \left. - \gamma^2 d_{ik}^T P_{ii} d_{ik} - \gamma^2 \sum_{j \in N_i} d_{jk}^T P_{ij} d_{jk} \right) \\ & + \hat{W}_i \frac{\partial \psi(\delta_{i(k+1)})}{\partial \delta_{i(k+1)}} (\delta_{ik} + (d_i + c_i) g_i u_{ik} \\ & - \sum_{j \in N_i} a_{ij} g_j u_{jk} + (d_i + c_i) h_i d_{ik} - \sum_{j \in N_i} a_{ij} h_j d_{jk}) \\ & = \hat{W}_i^T (\phi(\delta_{ik}) - \phi(\delta_{i(k+1)})) \\ & - \frac{1}{2} \left(\delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} + \sum_{j \in N_i} u_{jk}^T R_{ij} u_{jk} \right. \\ & \left. - \gamma^2 \sum_{j \in N_i} d_{jk}^T P_{ij} d_{jk} - \gamma^2 d_{ik}^T P_{ii} d_{ik} \right. \\ & \left. - \gamma^2 \sum_{j \in N_i} d_{jk}^T P_{ij} d_{jk} \right) \end{aligned} \quad (38)$$

To minimize the residual square, find $\hat{W}_i$ such that:

$$E_1 = \frac{1}{2} e_{H_i}^2 \quad (39)$$

The weight update law for evaluating the network using the gradient descent method is:

$$\hat{W}_i^{n+1} = \hat{W}_i^n - \alpha_m \frac{\partial E_1}{\partial \hat{W}_i^n} \quad (40)$$

Where $\alpha_m > 0$ is the evaluation network learning rate.

The design of the execution and perturbation networks is similar to the design of the evaluation network and will not be repeated here; the specific design can be found in the reference [28].

4.3. Convergence analysis of the ADP algorithm

Theorem 4.1 ((Monotonicity criterion)): Define the control decision sequence $\{u_{ik}\}$, the disturbance decision

sequence $\{d_{ik}\}$, and the value function sequence $\{V_{ik}\}$. For any δ_{ik} , if $V_{(i+1)k}(\delta_{ik}) \geq V_{ik}(\delta_{ik})$, then the value function $\{V_{ik}\}$ is a monotonically increasing sequence; if $V_{(i+1)k}(\delta_{ik}) \leq V_{ik}(\delta_{ik})$, then the value function $\{V_{ik}\}$ is a monotonically decreasing sequence.

Proof: Assume that for a given $V_{ik}(\delta_{ik})$, if there exists $V_1(\delta_{ik}) \geq V_0(\delta_{ik}) \geq 0$, and for $V_{ik}(\delta_{ik})$ assume that there exists $V_{(i+1)k}(\delta_{ik}) \geq V_{ik}(\delta_{ik})$, then we have:

$$\begin{aligned} V_{(i+1)k}(\delta_{ik}) &= \min_{u_{ik}} \max_{d_{ik}} \left\{ \delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} \right. \\ &\quad \left. - \gamma^2 d_{ik}^T P_{ii} d_{ik} + V_{(i+1)k}(\delta_{ik}) \right\} \\ &\geq \min_{u_{ik}} \max_{d_{ik}} \left\{ \delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} \right. \\ &\quad \left. - \gamma^2 d_{ik}^T P_{ii} d_{ik} + V_{ik}(\delta_{ik}) \right\} \\ &= V_{ik}(\delta_{ik}) \end{aligned} \quad (41)$$

Therefore, the value function $\{V_{ik}\}$ is a monotonically increasing sequence.

Similarly, assume that for a given $V_{ik}(\delta_{ik})$, if there exists $V_1(\delta_{ik}) \leq V_0(\delta_{ik})$, and for $V_{ik}(\delta_{ik})$ assume that there exists $V_{(i+1)k}(\delta_{ik}) \leq V_{ik}(\delta_{ik})$, then we have:

$$\begin{aligned} V_{(i+1)k}(\delta_{ik}) &= \min_{u_{ik}} \max_{d_{ik}} \left\{ \delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} \right. \\ &\quad \left. - \gamma^2 d_{ik}^T P_{ii} d_{ik} + V_{(i+1)k}(\delta_{ik}) \right\} \\ &\leq \min_{u_{ik}} \max_{d_{ik}} \left\{ \delta_{ik}^T Q_{ii} \delta_{ik} + u_{ik}^T R_{ii} u_{ik} \right. \\ &\quad \left. - \gamma^2 d_{ik}^T P_{ii} d_{ik} + V_{ik}(\delta_{ik}) \right\} \\ &= V_{ik}(\delta_{ik}) \end{aligned} \quad (42)$$

Therefore, the value function $\{V_{ik}\}$ is a monotonically decreasing sequence.

Next, the convergence of the ADP algorithm will be proved. ■

Assumption 4.1: There exist $0 < \alpha < \beta < \infty$ such that the inequality:

$$0 \leq \alpha V^*(\delta_{i(k+1)}) \leq \beta M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \quad (43)$$

established.

Assumption 4.2: There exist $0 \leq \rho \leq 1 \leq \theta \leq \infty$ such that the following inequality holds:

$$0 \leq \rho V^*(\delta_{ik}) \leq V_{0k}(\delta_{ik}) \leq \theta V^*(\delta_{ik}) \quad (44)$$

established. Here, $V_{0k}(\delta_{ik}) \geq 0$ and $V^*(\delta_{ik})$ are the initial value function and the optimal value function, respectively.

Theorem 4.2 ((Convergence criterion)): For a non-linear discrete - time system, the control input is u_{ik} ,

the disturbance input is d_{ik} , and the value function is $V_{ik}(\delta_{ik})$. The update rule is shown in Equation (32). If the following inequality holds:

$$\begin{aligned} &\left(1 + \frac{\rho - 1}{(1 + \beta^{-1})^i}\right) V^*(\delta_{ik}) \\ &\leq V_{ik}(\delta_{ik}) \leq \left(1 + \frac{\theta - 1}{(1 + \beta^{-1})^i}\right) V^*(\delta_{ik}) \end{aligned} \quad (45)$$

Then there is $V_\infty = V^*(\delta_{ik})$, $u_\infty = u^*(\delta_{ik})$, $d_\infty = d^*(\delta_{ik})$.

Proof: According to inequality (43) we have:

$$\beta M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) - \alpha V^*(\delta_{i(k+1)}) \geq 0 \quad (46)$$

According to Assumptions 4.1 and 4.2, it is known that $\frac{\theta-1}{\beta+1} \geq 0$ and $\frac{\rho-1}{\beta+1} \leq 0$. Combining with Equation (46), we can obtain:

$$\begin{cases} \frac{\theta-1}{\beta+1} (\beta M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) - \alpha V^*(\delta_{i(k+1)})) \geq 0 \\ \frac{\rho-1}{\beta+1} (\beta M_i(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) - \alpha V^*(\delta_{i(k+1)})) \leq 0 \end{cases} \quad (47)$$

Reference [29] proves the inequality (46). First, it proves that $V_{ik}(\delta_{ik}) \leq (1 + \frac{\theta-1}{(1+\beta^{-1})^i}) V^*(\delta_{ik})$.

Inequality (44) can be proved by applying mathematical induction, and the proof consists of two parts, starting with the left side.

The inequality holds when $i = 0$. Let $i = 1$, we have:

$$\begin{aligned} V_k(\delta_k) &= \min_{u_k} \max_{d_k} \left\{ M(\delta_k, u_k, u_{-k}, d_k, d_{-k}) \right. \\ &\quad \left. + \alpha V_0(\delta_{(k+1)}) \right\} \\ &\leq \min_{u_k} \max_{d_k} \left\{ M(\delta_k, u_k, u_{-k}, d_k, d_{-k}) \right. \\ &\quad \left. + \alpha \theta V^*(\delta_{(k+1)}) \right\} \\ &= \min_{u_k} \max_{d_k} \left\{ \left(1 + \frac{\theta - 1}{\beta + 1}\right) \right. \\ &\quad \left. M(\delta_k, u_k, u_{-k}, d_k, d_{-k}) \right. \\ &\quad \left. + \alpha \left(\theta - \frac{\theta - 1}{\beta + 1}\right) V^*(\delta_{(k+1)}) \right\} \\ &= \left(1 + \frac{\theta - 1}{\beta + 1}\right) V^*(\delta_{(k+1)}) \end{aligned} \quad (48)$$

Assume that when $i = j-1$, ($j = 1, 2, 3, \dots$), this conclusion still holds:

$$V_{j-1}(\delta_{ik}) \leq \left(1 + \frac{\theta - 1}{(1 + \beta^{-1})^{j-1}}\right) V^*(\delta_{ik}) \quad (49)$$

For $i = j$ it can be deduced that:

$$\begin{aligned} V_j(\delta_{ik}) &= \min_{u_{ik}} \max_{d_{ik}} \left\{ M(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \right. \\ &\quad \left. + \alpha V_{j-1}(\delta_{i(k+1)}) \right\} \end{aligned}$$

$$\begin{aligned}
&\leq \min_{u_{ik}} \max_{d_{ik}} \left\{ M(\delta_{ik}, u_{ik}, u_{-ik}, d_{ik}, d_{-ik}) \right. \\
&\quad \left. + \left(1 + \frac{\theta - 1}{(1 + \beta^{-1})^j} \right) \alpha V^*(\delta_{i(k+1)}) \right\} \\
&= 1 + \frac{(\theta - 1)\beta^j}{(1 + \beta)^j} \min_{u_{ik}} \max_{d_{ik}} \{ M(\delta_{ik}, u_{ik}, u_{-ik}, \\
&\quad d_{ik}, d_{-ik}) + \alpha V^*(\delta_{i(k+1)}) \} \\
&= \left(1 + \frac{\theta - 1}{(1 + \beta^{-1})^j} \right) V^*(\delta_{ik}) \quad (50)
\end{aligned}$$

From Equations (46) to (50), it is proved that $V_{ik}(\delta_{ik}) \leq (1 + \frac{\theta-1}{(1+\beta^{-1})^j}) V^*(\delta_{ik})$ holds. Using the same method, it can be proved that $(1 + \frac{\rho-1}{(1+\beta^{-1})^j}) V^*(\delta_{ik}) \leq V_{ik}(\delta_{ik})$. This will not be repeated here.

Finally, the convergence of the value function as $i \rightarrow \infty$ is proved. For $0 \leq \theta \leq \infty$, as $i \rightarrow \infty$, we have:

$$\begin{cases} \lim_{i \rightarrow \infty} \left(1 + \frac{\rho - 1}{(1 + \beta^{-1})^i} \right) V^*(\delta_{ik}) = V^*(\delta_{ik}) \\ \lim_{i \rightarrow \infty} \left(1 + \frac{\theta - 1}{(1 + \beta^{-1})^i} \right) V^*(\delta_{ik}) = V^*(\delta_{ik}) \end{cases} \quad (51)$$

Then,

$$V_\infty(\delta_{ik}) = V^*(\delta_{ik}) \quad (52)$$

Similarly, according to Equation (51), it can also be proved that $u_\infty = u^*(\delta_{ik})$ and $d_\infty = d^*(\delta_{ik})$. In summary, Theorem 4.2 is proved. ■

In this section, Theorem 4.1 demonstrates that the value function $V_i(\delta_i)$ is either non-increasing or non-decreasing during iterative updates. Furthermore, Theorem 4.2 proves that the value function is bounded within a finite range. These two properties—monotonicity and boundedness—ensure the convergence of $V_i(\delta_i)$, as a monotonic sequence constrained within finite bounds is guaranteed to converge.

4.4. Analysis of computational complexity

To assess the practicality of the proposed ADP algorithm, we analyze its computational complexity in terms of the key components: value function updates, neural network training, and iterative convergence. The analysis provides insight into the algorithm's efficiency and its trade-offs compared to traditional methods such as Sliding Mode Control (SMC).

Value Function Update Complexity: The iterative update of the value function $V_i(\delta_i)$ involves matrix operations and neural network evaluations. For a single UAV, the computational cost per iteration can be expressed as:

$$C_{\text{value}} = O(n \cdot m) \quad (53)$$

where n represents the state dimension and m the control input dimension. This cost arises from evaluating the Hamilton-Jacobi-Isaacs (HJI) equation to derive the optimal control and disturbance strategies.

Neural Network Training Complexity: The ADP algorithm employs a three-layer neural network to approximate the value function. Let: l be the number of layers in the network, p be the number of neurones per layer, k be the total number of training iterations.

The forward and backward propagation for training the neural network contributes a computational complexity of:

$$C_{\text{nn}} = O(k \cdot l \cdot p^2) \quad (54)$$

This complexity reflects the cost of adjusting weights and biases during backpropagation, where p^2 accounts for the dense connections between neurones in adjacent layers.

Overall Algorithm Complexity: The ADP algorithm requires T iterations to achieve convergence of the value function and control strategies. For a system with N UAVs, the overall complexity can be approximated as:

$$C_{\text{total}} = O(N \cdot T \cdot (n \cdot m + k \cdot l \cdot p^2)) \quad (55)$$

Here, T captures the iterative nature of the algorithm, while the terms $n \cdot m$ and $k \cdot l \cdot p^2$ represent the contributions from value function updates and neural network training, respectively.

Comparison with SMC: In contrast, the computational complexity of Sliding Mode Control (SMC) is primarily determined by the evaluation of control laws, which involves simpler algebraic operations. For N UAVs, the complexity of SMC can be expressed as:

$$C_{\text{smc}} = O(N \cdot n \cdot m) \quad (56)$$

In comparison, the ADP algorithm indeed exhibits higher computational complexity and demands significantly more computational resources. However, as demonstrated in the analysis of simulation results in Section 5.2, despite requiring more computational time per iteration, the ADP algorithm significantly outperforms the SMC algorithm in terms of trajectory tracking accuracy, control input smoothness, and robustness against disturbances. These performance enhancements are particularly evident in complex dynamic environments, enabling the ADP algorithm to effectively handle the nonlinear dynamics and external disturbances of multi-UAV systems.

Specifically, the simulation results reveal that the ADP algorithm achieves rapid convergence and precise control through iterative updates of the value function and control strategies. Its optimized control inputs not only reduce energy consumption but also eliminate the chattering effects commonly observed in SMC algorithms. Moreover, the learning mechanism of the ADP

algorithm allows it to adjust strategies in real time based on system states and environmental changes, demonstrating a high degree of adaptability and stability.

5. Simulations

5.1. Simulation parameter settings

To validate the effectiveness of the Adaptive Dynamic Programming (ADP) algorithm, which is based on zero-sum game theory, in multi-UAV systems, this section designs and conducts simulation experiments. The simulation primarily aims to verify the trajectory tracking performance and robustness of the ADP algorithm, and compares it with the Sliding Mode Control (SMC) algorithm to highlight its superiority.

In this section, the number of follower UAVs is set as $N = 4$, the total time $T = 60$ s, and the time step $\Delta t = 0.1$ s. The trajectory of the virtual leader UAV is designed as a three-dimensional sinusoidal function: $x_{0(k+1)} = [15 \sin(0.2t) \ 15 \cos(0.2t) \ -6 + 0.2t]^T$. The communication topology structure of the four follower UAVs is $A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$. The performance index weight matrices are $Q_{ii} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $R_{ii} = 1$, $P_{ii} = 1$, and the coefficient $\gamma = 1$. And set the initial spatial positions of each follower UAV as $x_{0k} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 \end{bmatrix}$, and the safe distance between each UAV is set as 1 m. According to the definition formula (2), the nonlinear functions of each UAV are set as $f(x_{ik}) = \begin{bmatrix} 0.8x_{ik}(1) + 0.2x_{ik}(2) + 0.1x_{ik}(3) \\ -0.5 \sin(0.4x_{ik}(1))x_{ik}(1) + 0.995x_{ik}(2) \\ 0.1x_{ik}(2) + 0.9x_{ik}(3) \end{bmatrix}$, $g_i = \begin{bmatrix} 0.1 \\ x_{ik}(2) \\ 0.05 \end{bmatrix}$, $h_i = \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \end{bmatrix}$. Among them, $x_{ik}(1)$ represents the state variable in the x -axis direction, $x_{ik}(2)$ represents the state variable in the y -axis direction, and $x_{ik}(3)$ represents the state variable in the z -axis direction. In order to simulate the environmental constraints, in addition to setting the fixed disturbance $h_i d_i(k)$, a random wind disturbance is added. The random wind disturbance is generated by $wind_disturbance = 0.1 \times randn(3, 1)$

and superimposed on the state update formula during each state update.

In this simulation, the input layer processes the UAV state variables, the hidden layer consists of six neurons, and finally, the output layer contains a single neurone. To ensure effective training and convergence, the weights of the neural network are initialized to small random values within the range $[-0.25, 0.25]$. This initialization avoids potential issues such as gradient vanishing or explosion during the early iterations of training. The hidden layer employs the Sigmoid activation function, defined as $\sigma(x) = \frac{1}{1+e^{-x}}$, which introduces nonlinearity essential for capturing the UAV system's complex dynamics. The neural network training process uses a learning rate α , initially set to 0.01, and dynamically adjusts it based on the observed error. Specifically, if the error exceeds a predefined threshold, the learning rate gradually increases, capped at a maximum of 0.1. Conversely, if the error remains below the threshold, the learning rate decreases to ensure stable learning, with a minimum value of 0.001. This adaptive adjustment balances the need for fast convergence and stability during training. The network weights are iteratively updated using the gradient descent algorithm. The error between the target value and the current network output propagates through the network, and the gradients of the activation function in the hidden layer are computed to update the weights. The weight updates follow the rule outlined in Equation (40), ensuring the network effectively learns the approximation of the value function.

5.2. Simulation results analysis

Figures 1, 2, and 3 present the tracking errors of the ADP and SMC algorithms along the X , Y , and Z axes, respectively. The results highlight the ADP algorithm's superior performance in achieving smaller tracking errors and faster convergence. For instance, in the X -axis (Figure 1), the steady-state tracking error of ADP

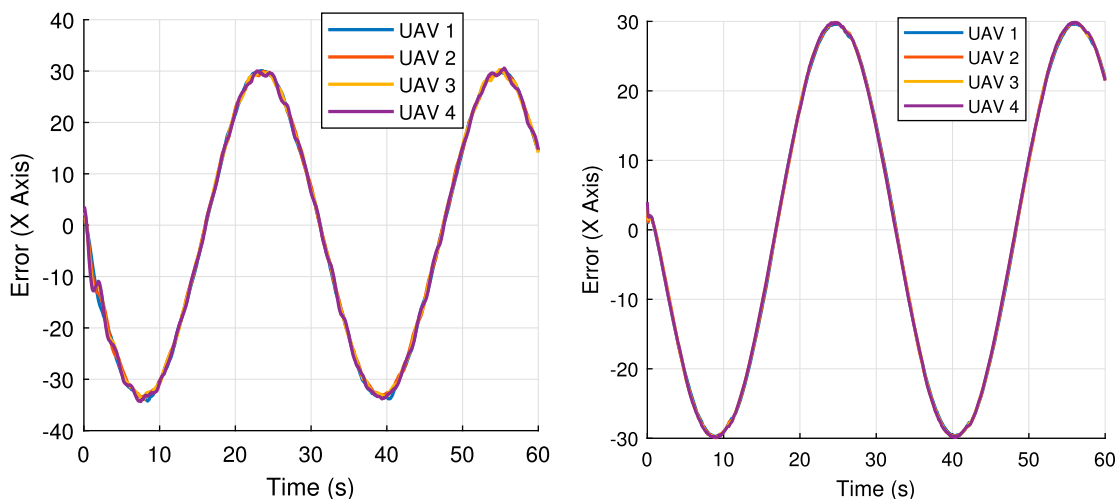


Figure 1. Tracking Error of UAV System in X-axis (ADP vs SMC).

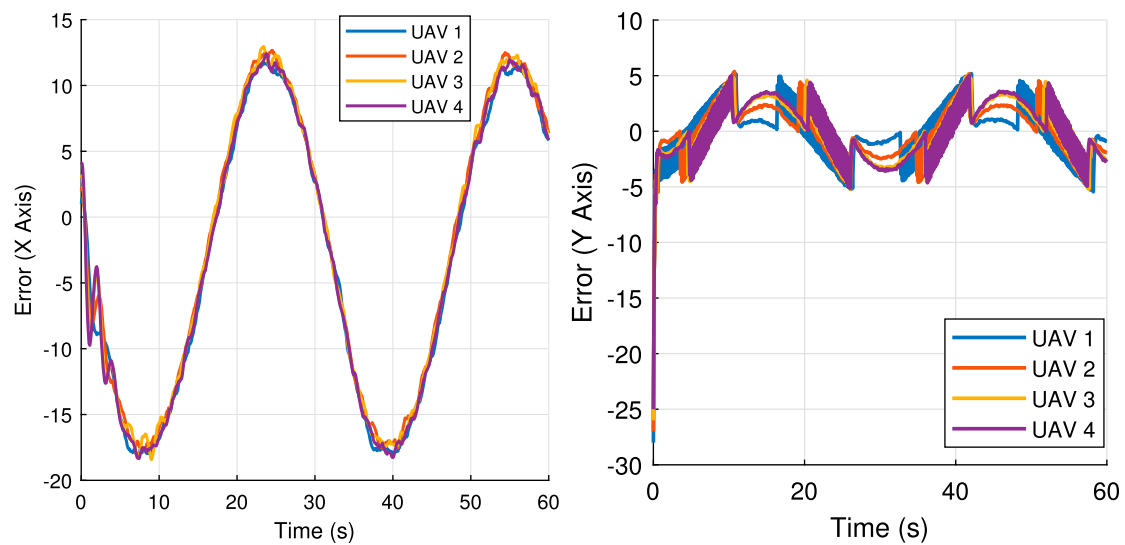


Figure 2. Tracking Error of UAV System in Y-axis(ADP vs SMC).

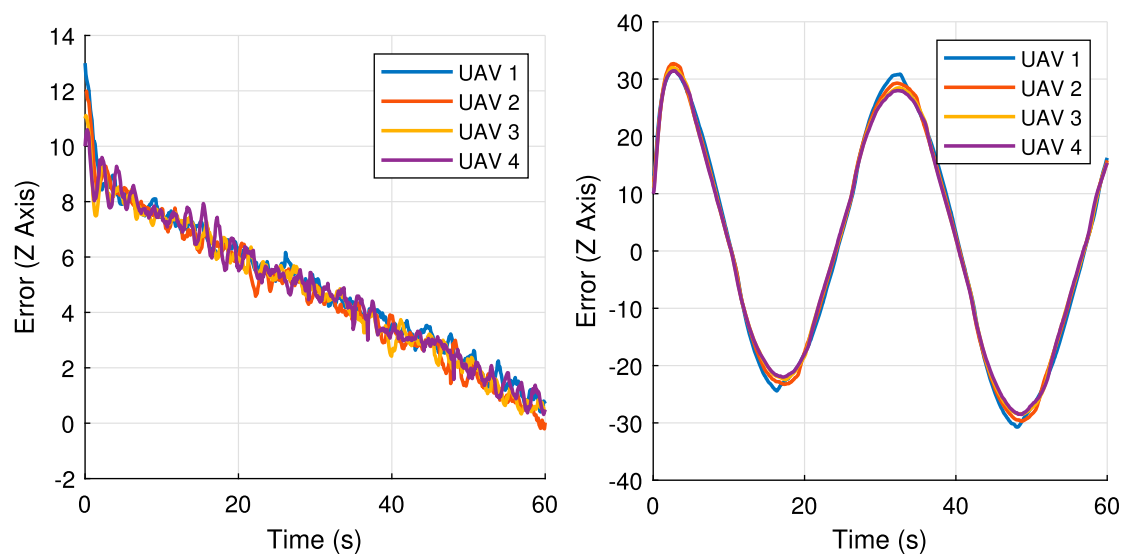


Figure 3. Tracking Error of UAV System in Z-axis(ADP vs SMC).

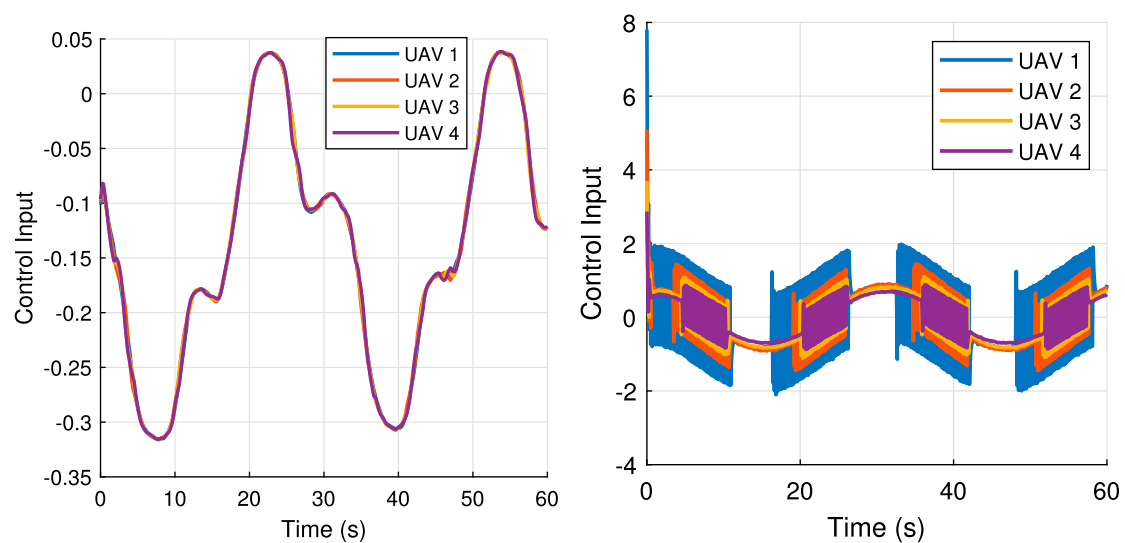


Figure 4. Control Inputs of UAV System(ADP vs SMC).

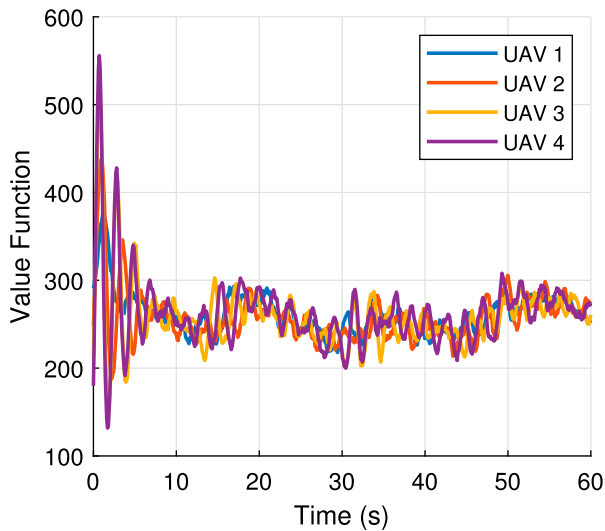


Figure 5. Value Function Curve of UAV System.

is approximately 0.05 units, compared to 0.15 units for SMC. A similar trend is observed in the Y-axis and Z-axis, where ADP demonstrates consistent advantages in precision and stability.

Figure 4 shows the control inputs generated by both algorithms. The ADP algorithm produces smoother and more stable control signals, effectively eliminating the chattering effects that are typical in SMC. This smoothness not only enhances control accuracy but also reduces energy consumption, making ADP more suitable for dynamic and complex environments.

Figure 5 illustrates the evolution of the value function under the ADP algorithm. The value function converges steadily over time, reflecting the efficiency of the neural network in approximating the optimal control strategy. This convergence indicates the algorithm's reliability in solving the Hamilton-Jacobi-Isaacs (HJI) equation.

Finally, Figure 6 presents the 3D trajectories of the UAVs under ADP control. The follower UAVs closely align with the leader's reference trajectory, with maximum deviations of 0.02 units in the X-axis and 0.03 units in the Z-axis. These results demonstrate the ADP algorithm's capability for precise trajectory tracking in complex 3D environments.

In summary, the ADP algorithm shows clear advantages over the SMC algorithm in tracking accuracy, control stability, and robustness against disturbances. These results validate the effectiveness of integrating zero-sum game theory with adaptive dynamic programming for multi-UAV systems in dynamic and complex scenarios.

6. Conclusions

In this study, we proposed an adaptive dynamic programming (ADP) optimal tracking control algorithm for multi-UAV systems based on zero-sum game

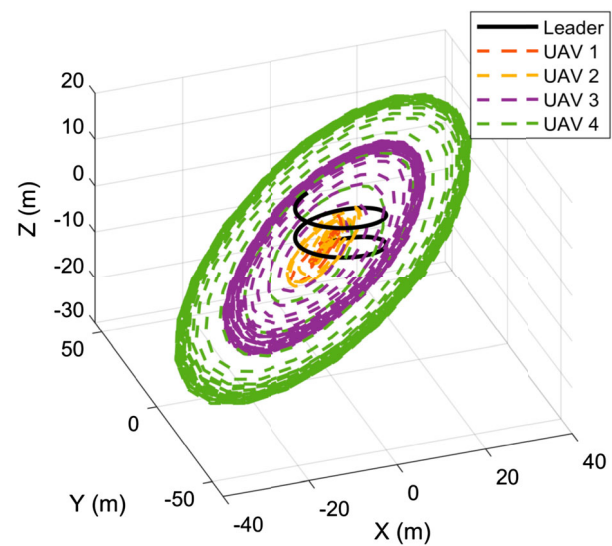


Figure 6. 3D Tracking Trajectory of UAV System.

theory. The performance index and Hamilton-Jacobi-Isaacs (HJI) equation were formulated to address the bounded L_2 gain problem, providing a rigorous framework for obtaining the Nash equilibrium solution. A three-layer neural network structure was designed to approximate the value function and iteratively optimize control strategies under the ADP framework, effectively solving the computational challenges associated with HJI equations.

The proposed algorithm was validated through comprehensive simulations, including comparisons with Sliding Mode Control (SMC). The results demonstrated that the ADP algorithm achieved superior trajectory tracking accuracy, smoother control inputs, and enhanced robustness against external disturbances. Furthermore, the adaptive learning and real-time optimization capabilities of the ADP algorithm highlighted its practical advantages in complex and dynamic multi-UAV environments.

These findings underscore the potential of integrating game theory with adaptive dynamic programming to develop robust and efficient control algorithms for multi-UAV systems. Future work will focus on incorporating more detailed UAV dynamics and environmental constraints into the proposed framework to further enhance its applicability and performance in real-world scenarios.

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References

- [1] Di B, Zhou R, Duan H. Potential field based receding horizon motion planning for centrality-aware multiple UAV cooperative surveillance. *Aerosp Sci Technol*. 2015;46:386–397. doi: [10.1016/j.ast.2015.08.006](#)
- [2] Zou Y, Jiang X, Yan P, et al. Next-generation unmanned aerial vehicle (UAV) cooperative communications. *Nat Sci Educ*. 2017;37:43–51.
- [3] Nash JF. Two-person cooperative games. *Econometrica*. 1950;21:128–140. doi: [10.2307/1906951](#)
- [4] Nash JF. Non-cooperative games. *Ann Math*. 1951;54:286–295. doi: [10.2307/1969529](#)
- [5] Starr AW, Ho YC. Nonzero-sum differential games. *J Optim Theory Appl*. 1969;3:184–206. doi: [10.1007/BF00929443](#)
- [6] Vamvoudakis KG, Lewis FL. Multi-player non-zero-sum games: online adaptive learning solution of coupled Hamilton-Jacobi equations. *Automatica*. 2011;47:1556–1569. doi: [10.1016/j.automatica.2011.03.005](#)
- [7] Siraj MS, Rahman AB, Diamanti M, et al. Alternative positioning, navigation, and timing enabled by games in satisfaction form and reconfigurable intelligent surfaces. *IEEE Syst J*. 2023;17(3):5035–5046. doi: [10.1109/JSYST.2023.3268989](#)doi: [10.1109/JSYST.2023.3268989](#)
- [8] Mu C, Wang D, He H. Novel iterative neural dynamic programming for data-based approximate optimal control design. *Automatica*. 2017;81:240–252. doi: [10.1016/j.automatica.2017.03.022](#)
- [9] Prokhorov DV, Santiago RA, Wunsch II DC. Adaptive critic designs: a case study for neurocontrol. *Neural Networks*. 1995;8(9):1367–1372. doi: [10.1016/0893-6080\(95\)00042-9](#)
- [10] Hendzel Z. An adaptive critic neural network for motion control of a wheeled mobile robot. *Nonlinear Dyn*. 2007;50(4):849–855. doi: [10.1007/s11071-007-9234-1](#)
- [11] Werbos P. A menu of designs for reinforcement learning over time. *Neural Networks*. 1990;3:67–95. doi: [10.1016/0893-6080\(90\)90088-3](#)
- [12] Wen-qian T, Hai-qin X, Yang L. Research on 3D path planning of UAV based on improved PSO hybrid algorithm. *J Qingdao Univ*. 2023;3:57–63.
- [13] Jingmin T, Bingwen W, Jiaqi H, et al. Trajectory optimization and resource allocation for full-duplex unmanned aerial vehicle-relaying. *J Beijing Univ Posts Telecommun*. 2024;47(1):51–57.
- [14] Guojie K, Shi F, Huilong Y, et al. A review on cooperative motion planning of unmanned vehicles. *Acta Armamentarii*. 2023;1:11–26.
- [15] Chao FU, Shanlin Y. Multi-UAV cooperative fight simulation system based on game theory. *J Syst Simul*. 2009;9:2591–2594.
- [16] Zhengyuan L, Yuanqing W, Yanzhou L, et al. Overview and prospect of multiple UAV swarms mission planning and formation flying. *J. Command Control*. 2023;6:623–636.
- [17] Te M, Zhiyuan Z, Junhu Z. Formation control of multiple UAVs and distributed game problem. *Mod Def Technol*. 2021;5:45–50+64.
- [18] Bo W, Pei-Yang Y, Fang-Chao J. Optimal self-organized cooperative control for mission rendezvous of multiple UAVs. *Electron Opt Control*. 2014;11:5–13.
- [19] Xiaoming Y. Research on multi UAV task based on cooperative game. *Electron Eng Product World*. 2016;7:46–47+42.
- [20] Yuan Y, Deng Y, Luo S, et al. Distributed game strategy for unmanned aerial vehicle formation with external disturbances and obstacles. *Front Inf Technol Electron Eng*. 2022;23(7):1020–1031. doi: [10.1631/FITEE.2100559](#)
- [21] Huifeng L, Wenfeng Y, Xiaoming C. Target tracking control algorithm based on approximate dynamic programming. *J Beijing Univ Aeronaut Astronaut*. 2019;3:597–605.
- [22] Wei Q, Liu D, Lin Q, et al. Adaptive dynamic programming for discrete-time zero-sum games. *IEEE Trans Neural Networks Learn Syst*. 2018;29(4):957–969.
- [23] Rizvi SAA, Lin Z. Output feedback adaptive dynamic programming for linear differential zero-sum games. *Automatica*. 2020;122:109272. doi: [10.1016/j.automatica.2020.109272](#)
- [24] Along W. Research on adaptive dynamic programming-based optimal cooperative control of UAV formation. Nanjing University of Aeronautics and Astronautics; 2018.
- [25] Lv Y, Ren X, Na J. Online optimal solutions for multi-player nonzero-sum game with completely unknown dynamics. *Neurocomputing*. 2018;283:87–97. doi: [10.1016/j.neucom.2017.12.045](#)
- [26] Di Z, Tingting L, Jiayou S. Optimal cooperative fault-tolerant control of UAV formation based on dynamic programming. *Electron Opt Control*. 2023;4:34–39.
- [27] Jiao Q, Modares H, Xu S, et al. Multi-agent zero-sum differential graphical games for disturbance rejection in distributed control. *Automatica*. 2016;69:24–34. doi: [10.1016/j.automatica.2016.02.002](#)
- [28] Ding W, Lingzhi H, Mingming Z, et al. Event-triggered control design for optimal tracking of unknown nonlinear zero-sum games. *Acta Autom Sin*. 2023;1:91–101.
- [29] Xiao G, Zhang H, Luo Y, et al. General value iteration based reinforcement learning for solving optimal tracking control problem of continuous-time affine nonlinear systems. *Neurocomputing*. 2017;245:114–123. doi: [10.1016/j.neucom.2017.03.038](#)